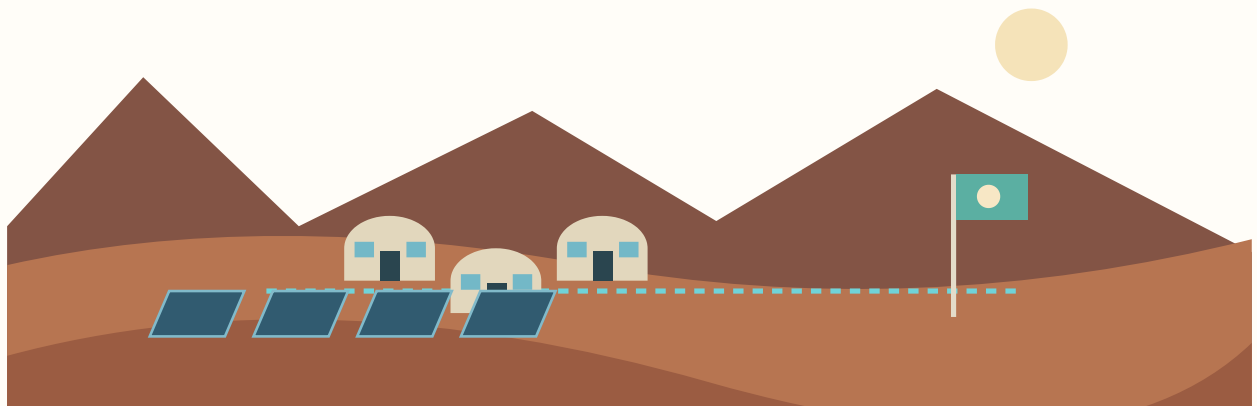


Mars Colony · Mission readiness report · v0.1.0

A WILLIAM MCADA PRODUCT · Pair Pair-07 · Resilient outpost · Teacher review of reasoning and independent work pending

MISSION DECISION

Resilient outpost



Story

The outpost is ready, with at least two days of each resource remaining.

Your task

Compare the final resource reserves and explain one model limitation to your teacher.

RED HORIZON

Pair-07

3W · 3O · 3H

4S · 1B

Mars Colony v0.1.0

System diagram · checked

System	Production →	→ Daily use	Storage
Water	90 L/day	72 L/day	144 L initially; 400 L cap
Oxygen	24 kg/day	19.2 kg/day	38.4 kg initially; 100 kg cap
Food	0 kg/day	14.4 kg/day	172.8 kg initially; 200 kg cap
Energy	144 normally / 108 storm kWh/day	114 kWh/day	36 kWh usable battery

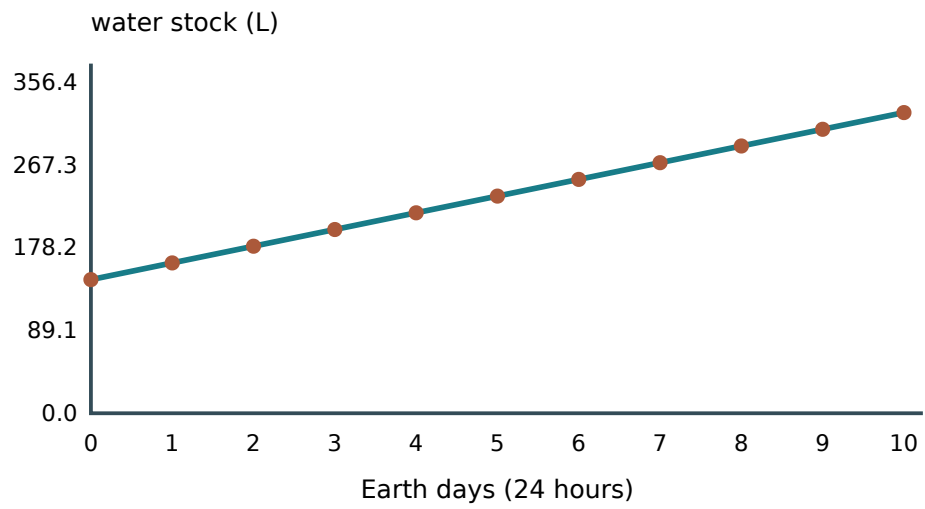
Production → daily use → storage change. Storage is an amount, not an extra daily production rate. This is daily accounting, not a model of continuous electrical supply.

Daily end-of-day stores; excess production spills at the cap

Earth day	Water (L)	Oxygen (kg)	Food (kg)	Battery (kWh)	Unserved energy (kWh)
0	144	38.4	172.8	36	0
1	162	43.2	158.4	36	0
2	180	48	144	36	0
3	198	52.8	129.6	36	0
4	216	57.6	115.2	30	0
5	234	62.4	100.8	24	0
6	252	67.2	86.4	18	0
7	270	72	72	36	0
8	288	76.8	57.6	36	0
9	306	81.6	43.2	36	0
10	324	86.4	28.8	36	0

Cargo manifest

Cargo item	Count / choice	Mass (kg)	Cost (cr)
water	3	450	4500
oxygen	3	600	6000
habitat	3	1200	9000
solar	4	400	4000
battery	1	80	800
Crate	standard	455.2	1000
TOTAL		3185.2	25300
REMAINING		214.8	700



Uncapped water stock rule: $S(t)=144 + (18)t$; whole Earth days 0–10. Daily storage capped; shortages flagged.

Learning evidence

Field	Task	First response	Current response	Status
waterDaily	Daily water demand	71	144/2	Checked
oxygenDaily	Daily oxygen demand	19.2	19.2	Checked
foodDaily	Daily food demand	14.4	14.4	Checked
waterTotal	Ten-day water demand	720	720	Checked
oxygenTotal	Ten-day oxygen demand	192	192	Checked
foodTotal	Ten-day food demand	144	144	Checked
liters	Classify L	stock	stock	Checked
waterRate	Classify L/day	rate	rate	Checked
kilograms	Classify kg	stock	stock	Checked
massRate	Classify kg/day	rate	rate	Checked
energy	Classify kWh	stock	stock	Checked
energyRate	Classify kWh/day	rate	rate	Checked
grams	Convert 0.8 kg to grams	800	800	Checked
milliliters	Convert 3 L to milliliters	3000	3000	Checked
minWater	Minimum water modules	3	3	Checked
minOxygen	Minimum oxygen modules	3	3	Checked
minHabitat	Minimum habitats	3	3	Checked
waterCapacity	Water capacity at the minimum count	90	90	Checked
oxygenCapacity	Oxygen capacity at the minimum count	24	24	Checked
housingCapacity	Occupancy at the minimum count	24	24	Checked
area	Selected habitat floor area	72	72	Checked
occupancy	Selected habitat occupancy	24	24	Checked
perPerson	Selected floor area per settler	3	3	Checked
mass	Total cargo	3185.2	3185.2	Checked
cost	Total cost	25300	25300	Checked
cargo	Cargo capacity remaining (may be negative)	214.8	214.8	Checked
budget	Budget remaining (may be negative)	700	700	Checked
initial	Stock equation: initial amount	144	144	Checked
net	Stock equation: signed net daily change	18	18	Checked
xAxis	Horizontal axis	Earth days	Earth days	Checked
yAxis	Vertical axis unit	L	L	Checked
table0	Table: stock after day 0	144	144	Checked
point0	Plot point: y-value at x = 0	144	144	Checked
table5	Table: stock after day 5	234	234	Checked
point5	Plot point: y-value at x = 5	234	234	Checked

Field	Task	First response	Current response	Status
table10	Table: stock after day 10	324	324	Checked
point10	Plot point: y-value at x = 10	324	324	Checked
percent	Percentage of solar yield remaining	75	75	Checked
perArray	Storm yield per array	27	27	Checked
load	Total daily electrical load	114	114	Checked
normal	Normal solar output	144	144	Checked
storm	Storm solar output	108	108	Checked
balance	Signed storm surplus / deficit	-6	-6	Checked
battery3	Stored energy at end of day 3	36	36	Checked
battery4	Stored energy at end of day 4	30	30	Checked
battery5	Stored energy at end of day 5	24	24	Checked
battery6	Stored energy at end of day 6	18	18	Checked
battery7	Stored energy at end of day 7	36	36	Checked
unserved	Total unserved energy during days 4-6	0	0	Checked
compareCost	Reference B minus A: cost	200	200	Checked
compareMass	Reference B minus A: cargo	20	20	Checked
compareBattery	Reference A: stored energy after day 6	18	18	Checked
compareSurplus	Reference B: storm daily surplus	21	21	Checked
transferAWater	Partner A: daily water for 30 settlers	90	90	Checked
transferAModules	Partner A: minimum 30 L/day water modules	3	3	Checked
transferBWater	Partner B: daily water for 28 settlers	84	84	Checked
transferBModules	Partner B: minimum 30 L/day water modules	3	3	Checked

Design defense — teacher review

Choice A: A costs 200 cr less; its battery has 18 kWh after day 6. B avoids the modeled deficit.

Limitation: The daily model ignores night-time generation and battery losses.

Help used: waterDaily. Model quantities are invented classroom assumptions.

Attempt history

Field	Response	Result	Scenario
waterDaily	71	Recheck	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
oxygenDaily	19.2	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
foodDaily	14.4	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
waterTotal	720	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
oxygenTotal	192	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
foodTotal	144	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
liters	stock	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
waterRate	rate	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
kilograms	stock	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
massRate	rate	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
energy	stock	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
energyRate	rate	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
grams	800	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
milliliters	3000	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
waterDaily	144/2	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
oxygenDaily	19.2	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
foodDaily	14.4	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
waterTotal	720	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
oxygenTotal	192	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
foodTotal	144	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
liters	stock	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
waterRate	rate	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
kilograms	stock	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
massRate	rate	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
energy	stock	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
energyRate	rate	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
grams	800	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
milliliters	3000	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
minWater	3	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
minOxygen	3	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
minHabitat	3	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
waterCapacity	90	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
oxygenCapacity	24	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
housingCapacity	24	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
area	72	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard

Field	Response	Result	Scenario
occupancy	24	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
perPerson	3	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
mass	3185.2	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
cost	25300	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
cargo	214.8	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
budget	700	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
initial	144	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
net	18	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
xAxis	Earth days	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
yAxis	L	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
table0	144	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
point0	144	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
table5	234	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
point5	234	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
table10	324	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
point10	324	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
initial	144	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
net	18	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
xAxis	Earth days	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
yAxis	L	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
table0	144	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
point0	144	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
table5	234	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
point5	234	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
table10	324	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
point10	324	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
percent	75	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
perArray	27	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
load	114	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
normal	144	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
storm	108	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
balance	-6	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
battery3	36	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
battery4	30	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
battery5	24	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
battery6	18	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard

Field	Response	Result	Scenario
battery7	36	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
unserved	0	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
compareCost	200	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
compareMass	20	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
compareBattery	18	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
compareSurplus	21	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
transferAWater	90	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
transferAModules	3	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
transferBWater	84	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard
transferBModules	3	Correct	water:3 oxygen:3 habitat:3 solar:4 battery:1 crate:standard

